

Regulatory Separation and Pharmaceutical Innovation: Evidence from China's MAH Reform

Abstract

This paper develops a partial-equilibrium mechanism model of how China's Marketing Authorization Holder (MAH) system changes the value of advancing viable drug projects and their commercialization organization. Developers choose one common original-drug innovation investment / project-advancement intensity before project characteristics and organizational routes are realized. MAH makes retained entrusted manufacturing legally available while the developer retains authorization-holder responsibility. Developers differ in project-advancement capability and internal manufacturing capability. Internal and entrusted production use distinct technologies and organizational costs. The new route is especially valuable when weak internal manufacturing capability makes vertical integration costly. Route choice remains deterministic, and the price of qualified manufacturing services is endogenous. Reform effects are confined to project-developer pairs for which the new route improves on the best old option; service scarcity can attenuate these gains. Financing frictions may amplify this organizational mechanism, but financing is not a baseline policy channel. The baseline does not treat MAH as a demand, credit-supply, scientific-productivity, downstream-success, or patenting shock.

Keywords: MAH; pharmaceutical innovation; commercialization frictions; complementary assets; entrusted production; organizational choice.

JEL codes: D22; L65; O31; O38; I18.

1 Introduction

China's MAH system separates the right to hold marketing authorization from a strict requirement of in-house production. This institutional change is potentially important for drug development because many research-originating firms can generate valuable projects but lack complete manufacturing capacity. The central mechanism studied here is therefore not demand expansion or scientific productivity. It is an expected-commercialization-value channel: when a firm anticipates that an advancing project can later use qualified

external manufacturing while the firm retains authorization, the expected value of advancing that project may increase. The reform can be particularly important when weak internal manufacturing capability makes vertical integration costly. A developer without suitable existing capacity can in principle build, acquire, validate, or adapt compliant production, but that internalization commitment can be large and front-loaded. The baseline captures this burden through route-specific setup and operating costs rather than modeling financial intermediation. A financing-friction extension is considered separately.

The mechanism builds on two established ideas. First, pharmaceutical innovation responds to expected downstream rewards ([Schmookler, 1966](#); [Acemoglu and Linn, 2004](#); [Finkelstein, 2004](#); [Dubois et al., 2015](#); [Budish et al., 2015](#)). Second, an innovator’s ability to appropriate returns depends on complementary assets, firm boundaries, and access to specialized inputs ([Teece, 1986](#); [Arora et al., 2001](#); [Gans and Stern, 2003](#); [Grossman and Helpman, 2002](#); [Bøler et al., 2015](#); [Ma, 2026](#)). MAH combines these ideas narrowly: it changes how an innovator can access manufacturing capability while retaining the authorization asset. It does not shift credit supply.

Recent evidence on China’s pharmaceutical reforms helps separate this route mechanism from two nearby channels. [Jia et al. \(2023\)](#) study the 2015 drug-review reform and emphasize approval delay, review capacity, and the composition of firms and products entering the pipeline. [Barwick et al. \(2026\)](#) study the National Reimbursement Drug List and emphasize demand and market-size incentives. Those channels can coexist with MAH, but they are not the MAH mechanism in this paper. The baseline holds scientific productivity and the market-return environment fixed and asks whether a regulated holder–producer separation route changes expected commercialization value.

Closely related evidence also shows why project advancement and upstream patenting must remain distinct. [Gu \(2024\)](#) documents heterogeneous trial and patent responses by financial resources. Those findings motivate a financing-friction extension and empirical heterogeneity tests. The baseline here isolates a different and more primitive MAH margin: the availability and value of retained entrusted manufacturing relative to internal production, transfer, and delay. This is a maintained model boundary, not an overstatement of novelty; the baseline neither generates patents nor treats financing as the policy channel.

For a development-economics audience, the relevant bottleneck is a regulated complementary asset. MAH can relax a manufacturing bottleneck for research-oriented firms while preserving holder responsibility. Scarcity in qualified production support can still attenuate the effect.

The model is deliberately narrow. It is not a full dynamic structural model of the pharmaceutical industry. It does not solve product-market clearing, an input market

for project advancement, free entry, exit, or an invariant firm distribution. The baseline closes only the market directly tied to the MAH mechanism: qualified manufacturing services. Reform-induced demand for retained entrusted production can raise the equilibrium service price p_m^* , so the model does not mechanically imply a positive effect for every project or developer.

The paper identifies a commercialization-organization channel of pharmaceutical innovation. By separating marketing-authorization ownership from physical manufacturing, MAH allows developers with weak internal manufacturing capability to retain commercialization rights while purchasing qualified production services. The model characterizes capability-based organizational sorting, the project-developer pairs for which the new route creates value, and the attenuation generated by scarcity in qualified manufacturing capacity. Financing frictions may amplify the mechanism but are not required for the baseline result.

The main text gives the shortest complete commercialization baseline. The technical appendix records primitives, payoff accounting, pricing derivations, deterministic sorting, project advancement, qualified-capacity equilibrium, comparative statics, observed outcomes, and explicitly separated inactive extensions, including commercialization financing.

2 Institutional Background

The relevant institutional point is timing. Under China’s drug-registration framework, a firm need not already be the final commercial holder when it begins to advance a viable project. The economic choice is forward-looking: a firm invests today while anticipating the manufacturing and commercialization environment it will face if a project advances. Production support and manufacturing readiness must be planned before final approval is observed. After obtaining a Drug Approval License, the applicant becomes the MAH ([National Medical Products Administration, 2022a](#)). The model therefore treats MAH as changing the anticipated availability and value of a future organizational route, not current scientific productivity or downstream success.

The MAH reform creates a regulated holder-producer separation route. The 2016 pilot allowed eligible applicants in selected regions to become holders under the pilot arrangement ([General Office of the State Council of the People’s Republic of China, 2016](#); [China Food and Drug Administration, 2016](#)). The later legal framework consolidated the principle that an MAH may manufacture by itself or entrust production to a qualified manufacturer ([National Medical Products Administration, 2019](#)). Subsequent NMPA rules require holder-side quality responsibility, assessment of the entrusted manufacturer, quality and entrustment agreements, and ongoing supervision ([National Medical Products](#)

Administration, 2022c,b, 2023).

These rules imply two modeling restrictions. First, entrusted production is not anonymous outsourcing or transfer. It is a regulated retained route in which authorization remains with the developer while production is performed by a qualified external manufacturer. Second, MAH does not eliminate holder responsibility. The entrusted payoff therefore retains a holder-side burden μ_E , while the institutional wedge $\tau_E(M)$ governs effective route availability. The downstream realization probability $s(q)$ is route independent and invariant to MAH in the baseline.

3 Model

3.1 Environment and timing

There is a continuum of developers. Developer i has predetermined project-advancement capability $a_i > 0$, internal manufacturing capability $k_i > 0$, with

$$\theta_i = (a_i, k_i) \sim H(a, k). \quad (1)$$

The joint distribution permits arbitrary correlation. A viable project that reaches commercialization-relevant planning draws value $q > 0$ and manufacturing requirement $m > 0$ from the exogenous distribution $F(q, m)$. For empirical decompositions only,

$$F(q, m) = \sum_{g \in \{O, \text{Inc}\}} \rho_g F_g(q, m), \quad \rho_g \geq 0, \quad \sum_g \rho_g = 1.$$

The class label g creates no class-specific control.

The institutional regime is $M \in \{0, 1\}$. The pre-MAH regime sets the barrier to retained entrusted manufacturing at infinity; the MAH regime makes it finite:

$$\tau_E(0) = +\infty, \quad \tau_E(1) = \bar{\tau}_E < +\infty, \quad \bar{\tau}_E \geq 0. \quad (2)$$

The available routes are internal manufacturing I , retained entrusted manufacturing E , non-retained transfer T , and abandonment or delay A . Under E , the developer remains the authorization holder and a qualified external producer manufactures. Thus E is not transfer.

Before project characteristics and routes are realized, developer i chooses one common $x_i \geq 0$, interpreted as original-drug innovation investment / project-advancement intensity. It is broader than pure clinical development, but excludes basic research, compound discovery, patent-generating effort, and patent applications. The expected measure of

viable projects reaching route planning is

$$\lambda_i^{\text{plan}} = a_i x_i. \quad (3)$$

Timing is as follows. First, M is observed and developers anticipate the market-consistent qualified-service price p_m^* . Second, after observing (a_i, k_i, M) , they choose x_i . Third, planning-stage projects draw (q, m) and choose among I, E, T, A . Fourth, downstream realization occurs with exogenous probability $s(q)$. Finally, aggregate capacity demand and supply must validate the anticipated service price. The last step is an equilibrium consistency condition, not a late unanticipated shock. Upstream research, patent generation, and financial intermediation are outside the baseline decision problem.

3.2 Commercialization technologies

Conditional on successful commercialization, residual demand is $y(p; q) = Aqp^{-\varepsilon}$, where $A > 0$ and $\varepsilon > 1$. Product-price optimization gives

$$p^*(c) = \frac{\varepsilon}{\varepsilon - 1} c$$

and the gross operating present value

$$R(q, c) = \frac{Aq}{1 - \beta\varphi} \frac{(\varepsilon - 1)^{\varepsilon-1}}{\varepsilon^\varepsilon} c^{1-\varepsilon}, \quad \beta \in (0, 1), \quad \varphi \in [0, 1). \quad (4)$$

The appendix derives the FOC, SOC, and accounting. In particular, $R_q > 0$ and $R_c < 0$.

Internal production uses marginal cost $c_I(m, k_i)$, with $c_{I,m} > 0$ and $c_{I,k} < 0$, and setup cost $F_I(m, k_i)$, with $F_{I,m} > 0$ and $F_{I,k} < 0$. Both are finite for every $k_i > 0$. A developer with little pre-existing manufacturing capability is not assumed to be permanently incapable of internal production. It can in principle build, acquire, validate, or adapt compliant capacity, but doing so is costly. Accordingly, $F_I(m, k_i)$ includes project-specific building, acquisition, validation, adaptation, production-readiness, quality-system, and organizational costs; it may be very large at low k_i . Its route value is

$$W_i^I(q, m) = s(q)R(q, c_I(m, k_i)) - F_I(m, k_i). \quad (5)$$

Entrusted production uses external marginal cost $c_E(m)$, requires $b(m) > 0$ units of qualified capacity with $b'(m) > 0$, incurs setup cost $F_E(m)$, and leaves the holder with burden $\mu_E \geq 0$. At the qualified-capacity price p_m ,

$$\begin{aligned} W_i^E(q, m; M, p_m) &= s(q)R(q, c_E(m)) - F_E(m) - p_m b(m) \\ &\quad - \mu_E - \tau_E(M). \end{aligned} \quad (6)$$

Entrusted production avoids establishing full proprietary manufacturing capability but still requires technology transfer, validation, contracted qualified capacity, and holder-side quality oversight. Each real monetary cost enters once. The transfer and abandonment values are

$$W^T(q, m) = T(q, m), \quad W^A = 0. \quad (7)$$

The baseline therefore compares real organizational costs and manufacturing capabilities without a route-financeability screen. Pre-MAH unavailability of E arises only from $\tau_E(0) = +\infty$. The appendix studies imperfect finance as an inactive amplification extension.

3.3 Organization and project advancement

At a fixed p_m , the project chooses deterministically:

$$\begin{aligned} W_i(q, m; M, p_m) &= \max\{W_i^I, W_i^E, W^T, W^A\}, \\ r_i^*(q, m; M, p_m) &\in \arg \max_{r \in \{I, E, T, A\}} W_i^r. \end{aligned} \quad (8)$$

Continuous heterogeneity makes ties a zero-measure event; no logit or inclusive value is used.

The internal-minus-entrusted gap is

$$\begin{aligned} \Delta_{IE}(k_i) &= s(q)[R(q, c_I(m, k_i)) - R(q, c_E(m))] - F_I(m, k_i) \\ &\quad + F_E(m) + p_m b(m) + \mu_E + \tau_E(M). \end{aligned} \quad (9)$$

Holding (q, m, M, p_m) fixed,

$$\Delta_{IE,k} = s(q)R_c(q, c_I) c_{I,k} - F_{I,k} > 0.$$

Proposition 1 (Organizational sorting). *Suppose Δ_{IE} is continuous and crosses zero between two capability levels. Because $\Delta_{IE,k} > 0$, there is a unique conditional cutoff $k^*(q, m; p_m, M)$ satisfying $\Delta_{IE}(k^*) = 0$. Developers below it select E , and those above it select I , conditional on both retained routes dominating T and A . If transfer or abandonment dominates, the cutoff does not determine the realized route.*

Developers with weaker existing manufacturing capability face higher internal variable and setup costs. Retained entrusted production is therefore relatively more valuable for them, although transfer or abandonment may still dominate both retained routes.

Define the best pre-MAH value and post-MAH value by

$$\begin{aligned} W_i^0(q, m) &= \max\{W_i^I(q, m), W^T(q, m), 0\}, \\ W_i^1(q, m; p_m) &= \max\{W_i^0(q, m), W_i^E(q, m; 1, p_m)\}. \end{aligned} \quad (10)$$

The MAH-relevant set is

$$\mathcal{C}_i(p_m) = \{(q, m) : W_i^E(q, m; 1, p_m) > W_i^0(q, m)\}. \quad (11)$$

Proposition 2 (MAH-relevant project set). *At a fixed p_m ,*

$$W_i^1 - W_i^0 = [W_i^E - W_i^0]_+, \quad (12)$$

so a strict project-value gain occurs exactly on $\mathcal{C}_i(p_m)$. Outside this set, the newly available route does not improve on the best old option and the direct project-value effect is zero.

Expected optimized value per planning-stage project is

$$\Omega_i(M, p_m) = \int W_i(q, m; M, p_m) dF(q, m). \quad (13)$$

Developer i solves

$$\max_{x_i \geq 0} \left\{ \beta a_i x_i \Omega_i(M, p_m) - \frac{\kappa}{1 + \nu} x_i^{1+\nu} \right\}, \quad \kappa > 0, \quad \nu > 0. \quad (14)$$

Strict concavity gives the unique solution

$$x_i^*(M, p_m) = \left[\frac{\beta a_i}{\kappa} \Omega_i(M, p_m) \right]^{1/\nu}, \quad (15)$$

including $x_i^* = 0$ when $\Omega_i = 0$. The binary reform affects x_i^* only through Ω_i . At a fixed p_m , its effect is strict only if E improves on the pre-MAH maximum on a positive-probability project set.

Proposition 3 (Project advancement and heterogeneity). *At a fixed p_m , the reform strictly raises x_i^* if and only if*

$$\int_{\mathcal{C}_i(p_m)} [W_i^E(q, m; 1, p_m) - W_i^0(q, m)] dF(q, m) > 0. \quad (16)$$

Advancement capability a_i scales any positive level response. Lower k_i raises internal variable and setup costs, making entrusted production relatively more valuable, but weak capability alone does not guarantee a treatment effect because T , A , or a low entrusted value may still dominate.

3.4 Qualified manufacturing-service equilibrium

Qualified supplier j , with efficiency z_j , chooses capacity $s_j \geq 0$:

$$\max_{s_j \geq 0} \{p_m s_j - \Psi(s_j; z_j)\}. \quad (17)$$

With $\Psi_{ss} > 0$, positive-price capacity is uniquely determined by $p_m = \Psi_s(s_j^*; z_j)$, and aggregate supply is

$$S_m(p_m) = \int s_j^*(p_m, z) dH_C(z).$$

Expected entrusted capacity per planning-stage project is

$$\chi_i^E(p_m; M) = \int b(m) \mathbf{1}\{r_i^*(q, m; M, p_m) = E\} dF(q, m). \quad (18)$$

Study-related and total demand are

$$\begin{aligned} D_m^{\text{MAH}}(p_m; M) &= \int a_i x_i^*(M, p_m) \chi_i^E(p_m; M) dH(a, k), \\ D_m(p_m; M) &= D_m^B(p_m) + D_m^{\text{MAH}}(p_m; M). \end{aligned} \quad (19)$$

The demand expression includes both feedbacks: price changes expected value and hence advancement, and it changes deterministic route selection. Continuous heterogeneity makes aggregate demand continuous even though individual indicators can jump.

The qualified-capacity market clears at the unique price

$$D_m(p_m^*; M) = S_m(p_m^*). \quad (20)$$

Supply is strictly increasing and demand weakly decreasing. Positive background demand gives excess demand at zero, while entrusted and background demand vanish and supply diverges at high prices. These conditions give existence and uniqueness. If post-MAH study demand is positive at the old price, $p_m^*(1) > p_m^*(0)$: CMO scarcity attenuates the reform rather than amplifying it through a direct price reduction.

For the equilibrium comparison, let $p_m^h \equiv p_m^*(h)$, and define

$$\Delta \Omega_i^{\text{dir}} \equiv \Omega_i(1, p_m^0) - \Omega_i(0, p_m^0), \quad \Delta \Omega_i^{\text{eq}} \equiv \Omega_i(1, p_m^1) - \Omega_i(0, p_m^0), \quad (21)$$

$$\Delta x_i^{\text{dir}} \equiv x_i^*(1, p_m^0) - x_i^*(0, p_m^0), \quad \Delta x_i^{\text{eq}} \equiv x_i^*(1, p_m^1) - x_i^*(0, p_m^0). \quad (22)$$

Proposition 4 (CMO equilibrium and scarcity attenuation). *Under atomless project and developer heterogeneity, the existing integrability conditions, weakly decreasing demand, and strictly increasing supply, the scalar CMO market has a unique clearing price.*

Moreover,

$$p_m^*(1) \geq p_m^*(0), \quad 0 \leq \Delta\Omega_i^{\text{eq}} \leq \Delta\Omega_i^{\text{dir}}, \quad 0 \leq \Delta x_i^{\text{eq}} \leq \Delta x_i^{\text{dir}}. \quad (23)$$

The price inequality is strict when post-MAH study demand is positive at the old price. MAH changes neither CMO supply nor the background demand schedule.

The baseline partial equilibrium contains exactly

$$\{p_m^*, x_i^*, r_i^*(q, m), s_j^*\}_{i,j}. \quad (24)$$

No entry, labor, capital, product-market aggregate, or welfare closure is added.

3.5 Observed outcomes and model boundary

Planning, organization, and observed outcomes. At equilibrium,

$$\Lambda_i^{\text{plan}} = a_i x_i^*,$$

while expected realized retained outcomes are

$$Y_i^{\text{ret}} = a_i x_i^* \int s(q) [\mathbf{1}\{r_i^* = I\} + \mathbf{1}\{r_i^* = E\}] dF(q, m). \quad (25)$$

Thus observed outcomes can change through advancement or route composition. Holder–producer separation is observed only after route assignment and cannot cause the earlier x_i choice.

Novelty composition. For $g \in \{O, \text{Inc}\}$, outcomes use the same x_i^* and the mixture $\rho_g F_g$. The baseline permits either class to have a zero or larger response and imposes no original-versus-incremental ranking. Patent applications are not a baseline endogenous outcome.

Financing extension boundary. The baseline does not model financial intermediation or impose a binding liquidity screen. Financing frictions may amplify the value of replacing a large internalization commitment with a contract-based route; the technical appendix analyzes that amplification separately and shows how the extension nests this baseline when liquidity never binds.

4 Empirical Predictions and Mapping Boundaries

The deterministic model yields three conditional predictions. First, among projects for which internal and entrusted production dominate transfer and abandonment, developers with weaker internal manufacturing capability are more likely to select retained entrusted production. Second, MAH raises project advancement only when the entrusted route exceeds the best pre-reform option; the response is zero when internalization, transfer, or delay remains weakly better. Third, reform-induced demand for qualified manufacturing services can raise p_m^* , attenuating fixed-price gains in expected project value and advancement. These predictions concern commercialization organization and project advancement, not patent generation or scientific productivity.

The empirical distinction is between ex ante advancement and downstream observations. Product-level holder–manufacturer separation can proxy route E only when holder identity, manufacturer identity, product identity, and effective date are aligned. Approval or launch records are realized-product outcomes, not direct observations of x_i . Clinical milestones may proxy project advancement only when their status and timing are measured consistently. Patent applications are a separate patenting margin and are not a baseline endogenous outcome. They may be studied as a separate empirical outcome, but the present mechanism neither maps them to x_i nor assigns their reform response a sign.

Table 1: Model Objects and Measurable Empirical Interfaces

Model object	Candidate interface	Claim boundary
Predetermined capability a_i Advancement x_i and $a_i x_i$	Pre-policy development history or other validated capability measure Consistently measured development-stage activity and planning-stage project events	Does not separately identify a_i , and no policy shift in a_i is allowed. x_i is not pure clinical effort, patent applications, or an approval count.
Internal manufacturing capability k_i	Pre-policy manufacturing assets, production licences, manufacturing history, validated technical scope, or other capability measures	This is the primary baseline heterogeneity; licence roles alone do not identify project-specific capability or realized route choice.
Route $r_i^* = E$	Product-level holder differs from manufacturer, with time-consistent identifiers	The observed split is downstream evidence on retained organizational assignment, not the policy treatment itself.
Retained outcome Y_i^{ret}	Approval or launch with developer/holder identity	Combines advancement, project mix, deterministic route assignment, and $s(q)$; it does not identify these primitives separately.
CMO scarcity p_m^*	Service price, or consistently measured capacity, utilization, density, waiting-time, or participation proxy	A nonprice proxy does not identify the price or the supply and demand schedules separately.
Novelty class g	Versioned regulatory classification O or Inc	The classifier creates no x_{ig} and imposes no response ranking.

The timing restriction can be summarized as

$$M \longrightarrow \text{anticipated availability and value of } E \longrightarrow \Omega_i \longrightarrow x_i^*, \quad (26)$$

followed by

$$\begin{aligned} x_i^* &\longrightarrow \text{planning-stage projects} \longrightarrow r_i^* \\ &\longrightarrow \text{observed holder-producer separation} \longrightarrow \text{realized products.} \end{aligned} \quad (27)$$

Observed separation cannot be placed before or treated as a cause of the ex ante x_i choice. The original and incremental classes use the same x_i and the mixture $\rho_g F_g$; either class may have a zero or larger response.

These entries are interfaces for future validated data. No data set is asserted here to be available, complete, or identifying. Before use, a source must pass provenance, coverage, key-uniqueness, unit, timing, and construct-validity checks. Entry, route-specific realization, and smooth route choice remain undeveloped extensions.

4.1 Extension-Based Heterogeneity: Financing Capacity

Financing is a secondary, extension-based heterogeneity test rather than a baseline identifying restriction. Candidate measures must predate reform exposure and may include registered or paid-in capital, cash and liquid assets, leverage or debt capacity, listed status, VC/PE backing, parent-company backing, or prior external financing. The empirical sequence is to establish first whether weak internal manufacturing capability predicts post-MAH retained entrusted production, then whether that route predicts project advancement or retained realization, and only afterward whether pre-policy financing remains informative after controlling for capability, size, age, ownership, group affiliation, listed status, and innovation history. A triple interaction among MAH exposure, weak manufacturing capability, and financing capacity is therefore conditional and secondary, not the model's primary baseline prediction. [Hall and Lerner \(2010\)](#) motivates the plausibility of this extension; MAH is not modeled as a credit-supply shock.

5 Conclusion

MAH changes expected project value by making retained entrusted manufacturing available while preserving authorization-holder responsibility. The route is most relevant when a developer can advance valuable projects but lacks project-compatible internal manufacturing capability. Weak capability raises the real variable and setup costs of internalization; the new retained route can replace that large, lumpy requirement with contracted qualified production. Its equilibrium value also depends on qualified-service

scarcity: reform-induced entrusted demand can raise p_m^* and attenuate the fixed-price gain. Project advancement rises only for the MAH-relevant set and need not rise for every developer or novelty class. Observed holder–producer separation and realized products occur after advancement and organizational assignment. Financing frictions can strengthen the mechanism when large internalization commitments are difficult to fund, but the baseline does not require a binding financing constraint. Patent applications and upstream scientific research remain outside the baseline, and the model makes no welfare claim.

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